



CVL2001: Climate Change and Adaptation Section B

Lecture 6: Climate Adaptation in Indian Cities
Ahmedabad Extreme Heat and Mumbai Urban Flooding

Instructor: Dr. Joseph V. Puthussery
Department of Civil and Environmental Engineering
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Learning Outcomes

1. Describe the 2010 Ahmedabad heatwave
2. Describe the 2005 Mumbai flood
3. Identify the climate risk in each city
4. Examine how each city can reduce harm

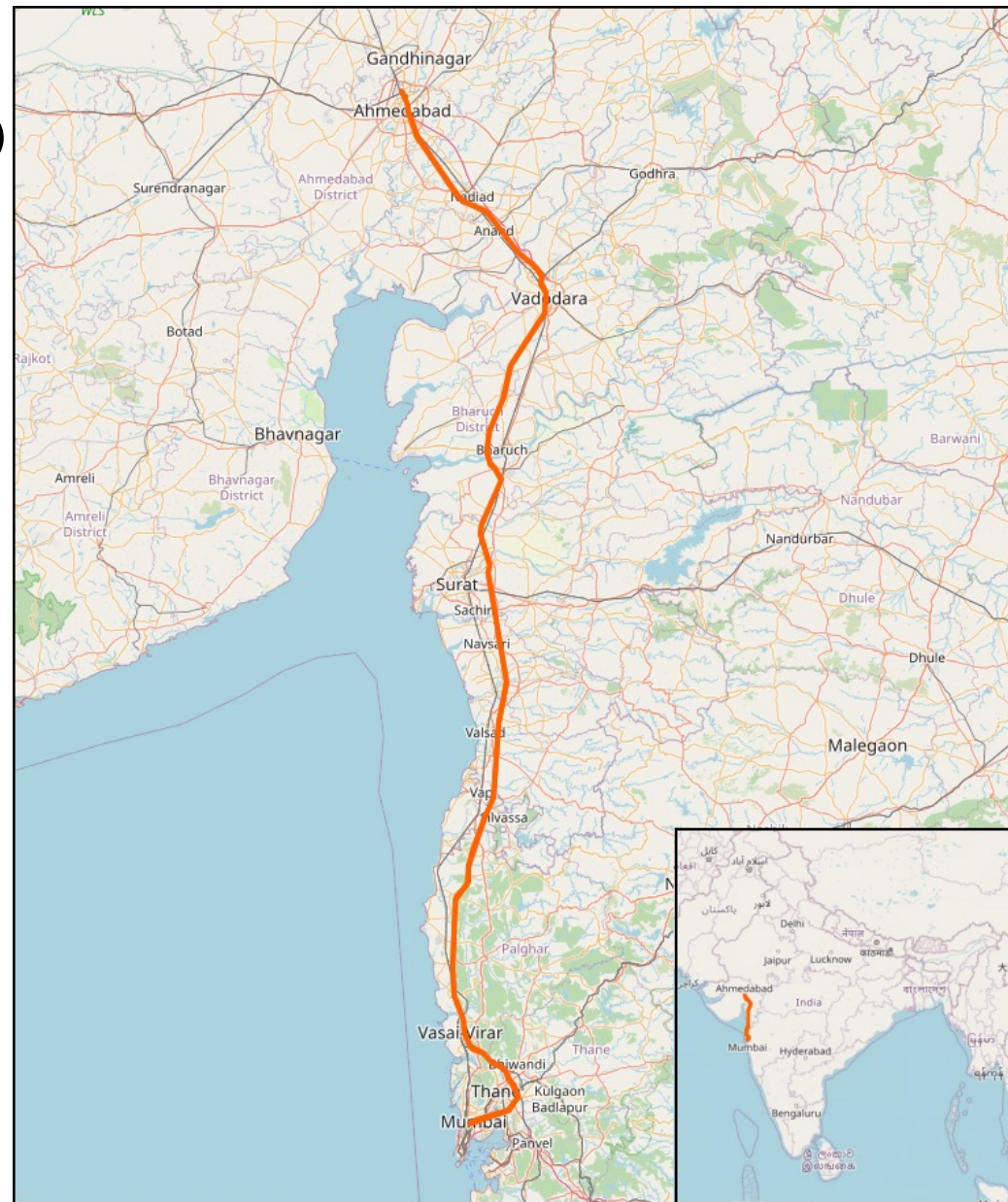
Two Cities, Two Climate Risks

Ahmedabad

- extreme heat
- risks to health and housing

Mumbai

- extreme rainfall and flooding
- risks to people, infrastructure and the economy



Case Study 1: Ahmedabad Heatwave 2010

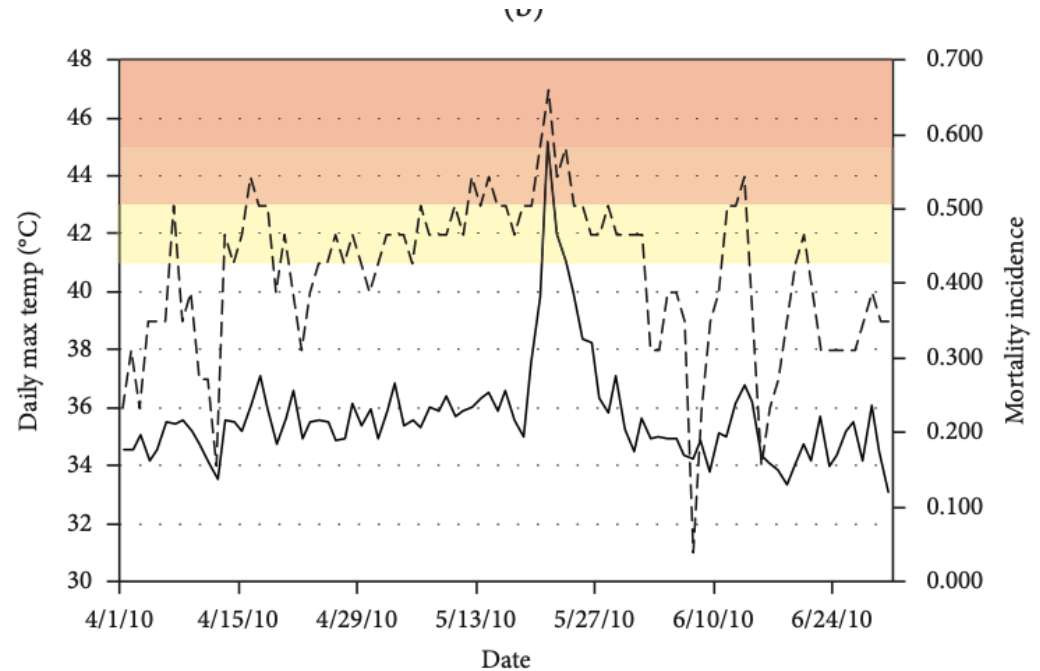
- A rapidly growing city with very hot summers
- Dense neighbourhoods increase heat exposure
- Many residents work outdoors or live in heat-trapping homes
- Climate risk reflects both temperature and vulnerability.
- Reference:
 - J. J. Hess *et al.*, Building resilience to climate change: Pilot evaluation of the impact of India's first heat action plan on all-cause mortality. *J. Environ. Public Health* **2018**, 7973519 (2018).
 - S. Vellingiri *et al.*, Combating climate change-induced heat stress: Assessing cool roofs and its impact on the indoor ambient temperature of households in the urban slums of Ahmedabad. *Indian J. Occup. Environ. Med.* **24**, 25–29 (2020).



<https://images.pexels.com/photos/33305052/pexels-photo-33305052.jpeg?cs=srgb&dl=pexels-arthsawale-33305052.jpg&fm=jpg>

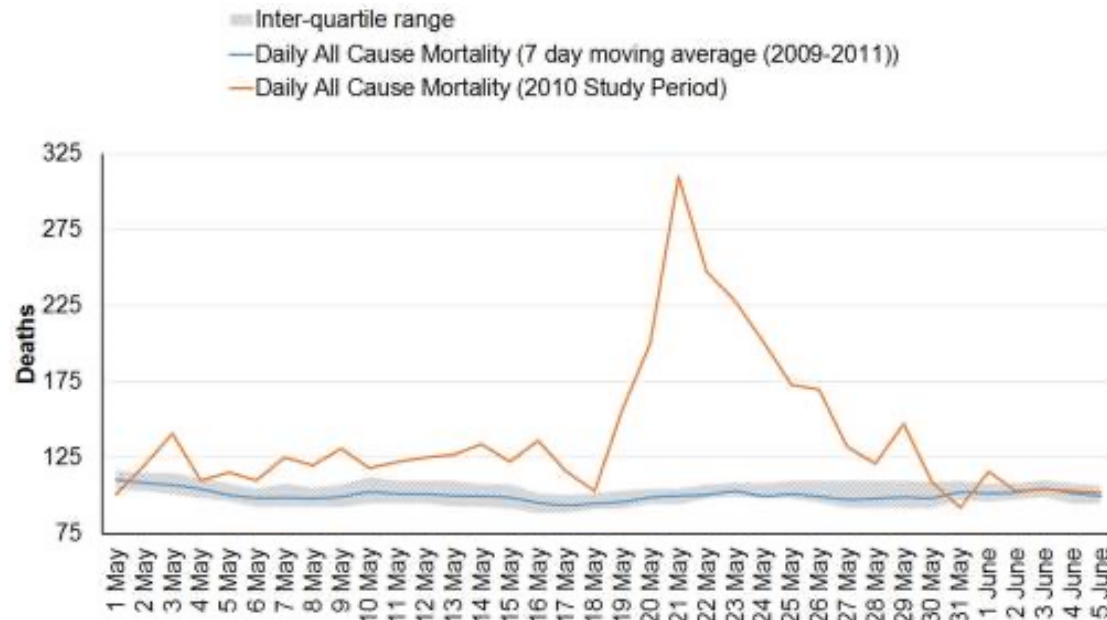
Ahmedabad: May 2010 Heatwave?

- Daily maximum temperature reached 46.8°C , the highest in nearly a century.
- Ahmedabad recorded $\sim 1,344$ excess deaths in May.
- Mortality was 43.1% above the baseline rate.
- Mortality increased rapidly above approximately 40°C .
- At 47°C , same-day mortality risk was 2.34 times the risk at 40°C .
- The event led to the development of Ahmedabad's Heat Action Plan.
- Reference: Hess *et al.*, *J. Environ. Public Health* 2018, 7973519 (2018).



Summer daily temperatures (dotted lines) and daily mortality incidence (solid lines). Deaths per 100,000 people per day

Ahmedabad: May 2010 Heatwave?



<https://www.ncbi.nlm.nih.gov/corecgit/tileshop/tileshop.fcgi?p=PMC3&id=994353&s=42&r=1&c=1>



<https://climahealth.info/wp-content/uploads/2023/03/Picture-1.png>

Ahmedabad: Why Was the Heat So Harmful?

- Mortality rises rapidly above a temperature threshold
- Outdoor work increases exposure
- Age and illness increase sensitivity
- Tin, cement-sheet and tarpaulin roofs trap heat
- The same temperature does not create equal risk.
- Reference: Hess et al. (2018); Vellingiri et al.



https://i1.wp.com/asiatimes.com/wp-content/uploads/2016/12/000_Del6451532-scaled.jpg?fit=2560%2C1679&ssl=1

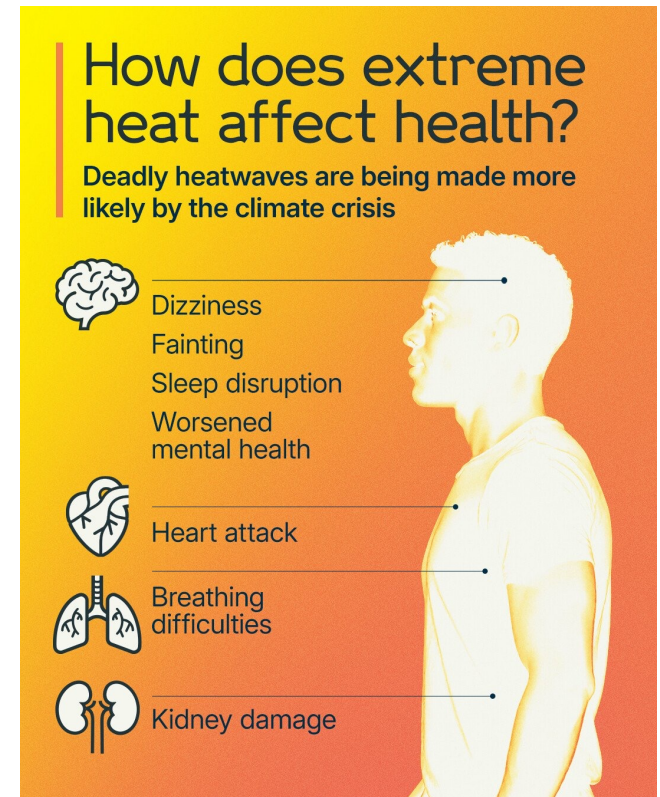
How Can Extreme Heat Cause Death?

The body normally releases heat through sweating and increased blood flow to the skin.

During extreme heat:

- **Heat gained > heat released**
 - core body temperature rises
 - dehydration reduces blood volume
 - the heart works harder to circulate blood
 - the brain, kidneys and other organs are damaged
 - heatstroke, organ failure and death
- Death occurs when the body can no longer maintain a safe internal temperature.

Source: WHO, *Heat and Health*.



https://cms.wellcome.org/sites/default/files/styles/image_full_hi/public/2025-06/MS-7383_22%20Heat%20Health%20Explainer_update_1080x1350.jpg?itok=4XrXor4D

Ahmedabad: Extreme Heat and Climate Change

- Climate warming shifts the temperature distribution
- Very hot days become more frequent or intense
- More days may cross health-protection thresholds
- A small rise in temperature can produce a much larger health impact.
- Reference: Hess et al. (2018)

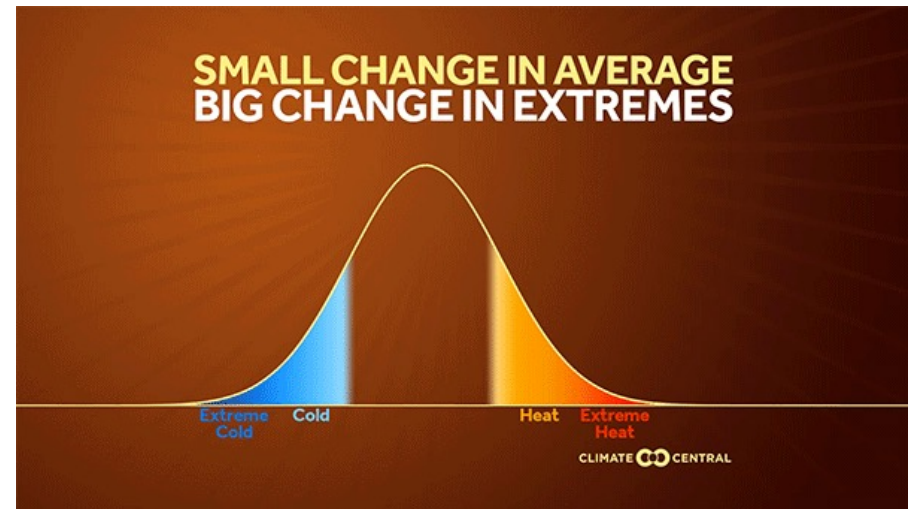


IMAGE PLACEHOLDER — SHIFTING TEMPERATURE DISTRIBUTION

Ahmedabad: What Did the Heat Action Plan Include?

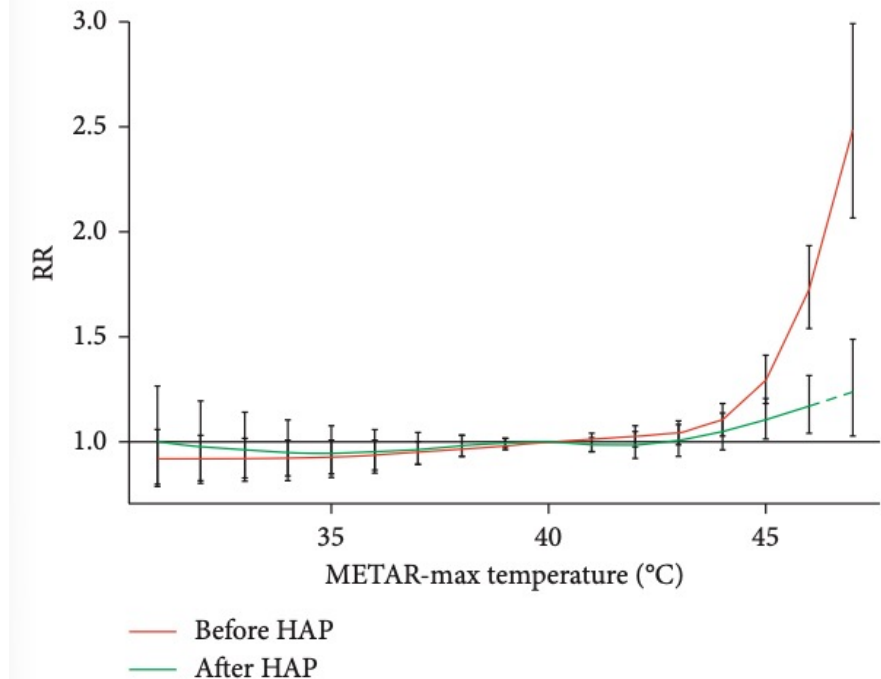
- A forecast-based alert system using daily maximum temperature:
 - **Yellow:** 41–42.9°C
 - **Orange:** 43–44.9°C
 - **Red:** ≥45°C
- Increasing levels of preparedness and response at each alert level.
- Public messages to reduce heat exposure, remain hydrated and use cooling measures.
- Training and preparedness for ambulances, clinics and hospital emergency departments.
- Coordination through a designated Ahmedabad Municipal Corporation health officer.
- Alert thresholds were derived from the observed temperature-mortality relationship.



Ahmedabad: Did the Plan Reduce Harm?

After implementation:

- At 47°C, the mortality risk ratio decreased from 2.34 before the plan to 1.25 after implementation.
- The study estimated approximately 1,190 avoided deaths per year during 2014–2015.
- Mortality reductions were greatest during the hottest days and highest alert levels.
- The findings suggest that heat warnings and preparedness improved Ahmedabad's resilience to extreme heat.
- Reference: Hess et al. (2018)



Ahmedabad: Did the Plan Reduce Harm?

TABLE 2: Counts of post-HAP days in each warning category and expected and observed mortality post-HAP and avoided mortality with 95% confidence intervals.

Variable	Any warning	Yellow warning	Orange warning	Red warning
Days Post-HAP	98	36	41	21
Expected deaths after HAP (95% CI)	16,012 (13,956–18,067)	5015 (4868–5162)	6307 (6114–6500)	4690 (2974–6406)
Observed deaths after HAP	13,632	4608	5634	3390
Avoided mortality after HAP (95% CI)	2380 (324–4435)	407 (260–554)	673 (480–886)	1300 (-416–3016)
Average annualized avoided mortality after HAP (95% CI)	1190 (162–2218)	203 (130–277)	336 (240–443)	650 (-208–1508)

After implementation:

- Mortality reductions were greatest during the hottest days and highest alert levels.
- The findings suggest that heat warnings and preparedness improved Ahmedabad's resilience to extreme heat.

Reference: Hess et al. (2018)

Ahmedabad: Household-Level Adaptation

Cool-roof options studied included:

- Reflective white paint
- Thermocol insulation
- Modular roofing systems
- These measures reduced indoor heat compared with untreated roofs.
- Reference: Vellingiri et al.



Figure 1: Four intervention type roofs: (clockwise) (a) Thermocol ceiling, (b) Modroof, (c) Solar reflective white paint, (d) Airlite ventilation

Ahmedabad: Household-Level Adaptation

Cool-roof performance

Reference: Vellingiri et al.

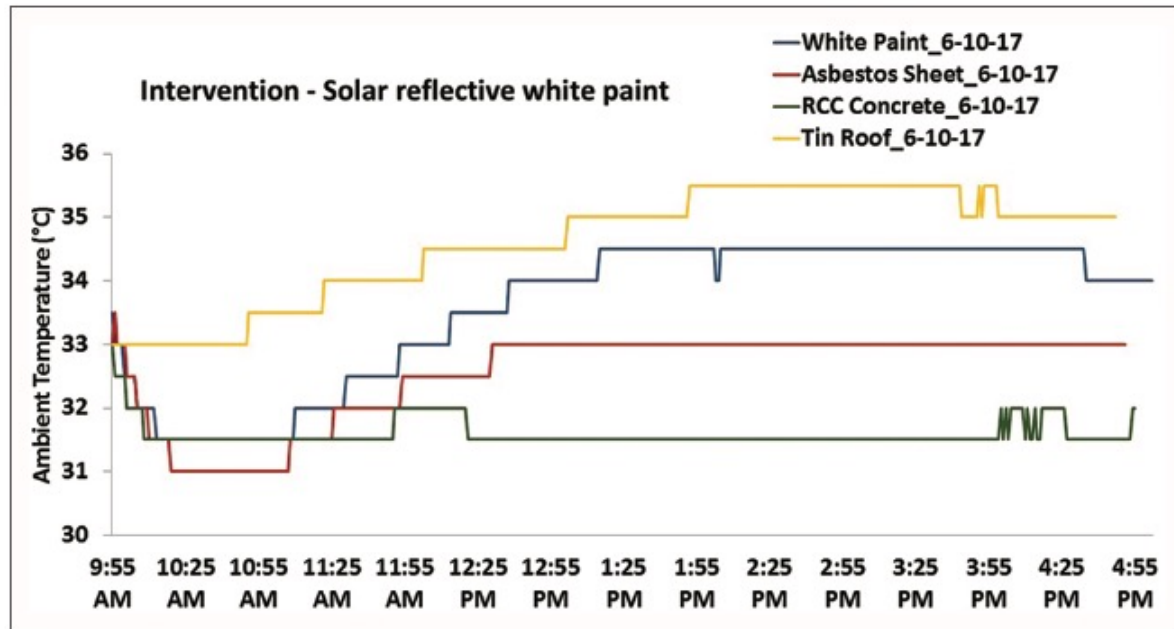


Figure 2: Plot of ambient temperature (°C) of solar reflective white paint and nonintervention roofs

Ahmedabad: Interpreting the Evidence

- The study compared pre-HAP summers (2007–2010) with post-HAP summers (2014–2015).
- Mortality at high temperatures was lower after implementation.
- However, this was an observational before–after study, not a randomized experiment.
- Changes in weather, behavior, healthcare, housing or population may also have influenced mortality.
- The results support the plan’s effectiveness, but do not prove that the HAP caused the entire reduction.

Reference: Hess *et al.*, *J. Environ. Public Health* **2018**, 7973519 (2018).

Q) How Can a Heat Warning Reduce Mortality?

- A. It prevents the heatwave from occurring
- B. It lowers the city's outdoor temperature
- C. It improves preparedness and reduces vulnerability
- D. It removes all exposure to extreme heat

Case Study 2 : Mumbai Floods 2005

- A densely populated coastal megacity
- Heavy monsoon rainfall is common
- Rivers, drains and low-lying areas carry excess water
- High tides can restrict drainage to the sea
- Flood risk combines natural and urban conditions.

Reference: Ranger et al. (2011)



<https://cdn.britannica.com/26/84526-050-45452C37/Gateway-monument-India-entrance-Mumbai-Harbour-coast.jpg>

Mumbai: What Happened in 2005?

- 944 mm of rain fell in 24 hours
- Flooding disrupted much of the city
- Around 500 fatalities were reported
- Direct damage was estimated near US\$2 billion
- The event revealed the scale of Mumbai's flood exposure.



https://th-i.thgim.com/public/migration_catalog/article14445109.ece/alternates/BASE_SQUARE/Mumbai

Reference: N. Ranger et al., An assessment of the potential impact of climate change on flood risk in Mumbai. Climatic Change 104, 139–167 (2011)

Mumbai: What Happened in 2005?

- **2:30 p.m., 26 July:** Extreme rainfall began over northern Mumbai.
- **Next three hours:** Approximately 381 mm of rain fell—about 127 mm/hour.
- **Late afternoon:** Heavy rainfall coincided with high tide, restricting drainage to the sea.
- **Within 3–4 hours:** Roads, railways and low-lying neighborhoods were inundated.
- **8:30 a.m., 27 July:** Santacruz recorded 944 mm in 24 hours; Colaba recorded only 74 mm.
- **Following 36 hours:** Transport gradually resumed, but disruption continued for several days.
- **Extreme rainfall + limited drainage + high tide → rapid urban flooding**



https://th-i.thgim.com/public/migration_catalog/article14445109.ece/alternates/BASE_SQUARE/Mumbai

Reference: N. Ranger et al., An assessment of the potential impact of climate change on flood risk in Mumbai. Climatic Change 104, 139–167 (2011)

Mumbai: Why Was the Flood So Destructive?

1. Extreme rainfall

→ exceeded drainage capacity

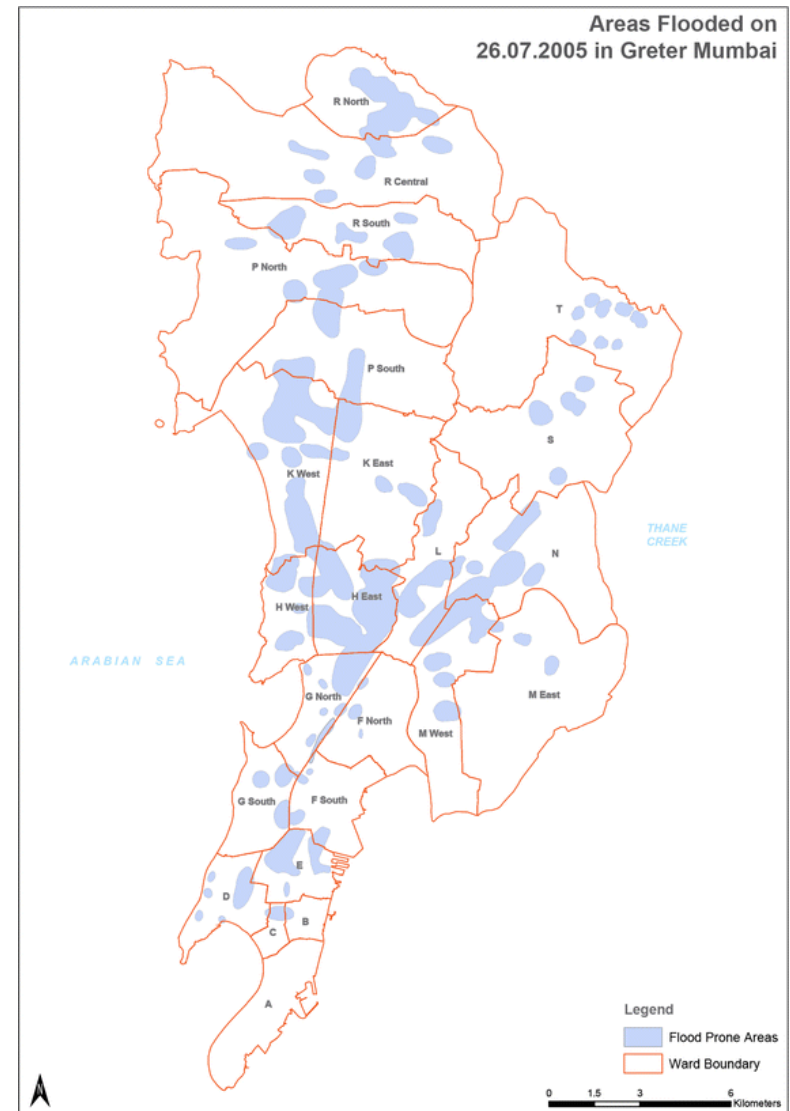
2. Dense development

→ placed more people and assets in harm's way

3. High tide and blocked channels

→ slowed the movement of water

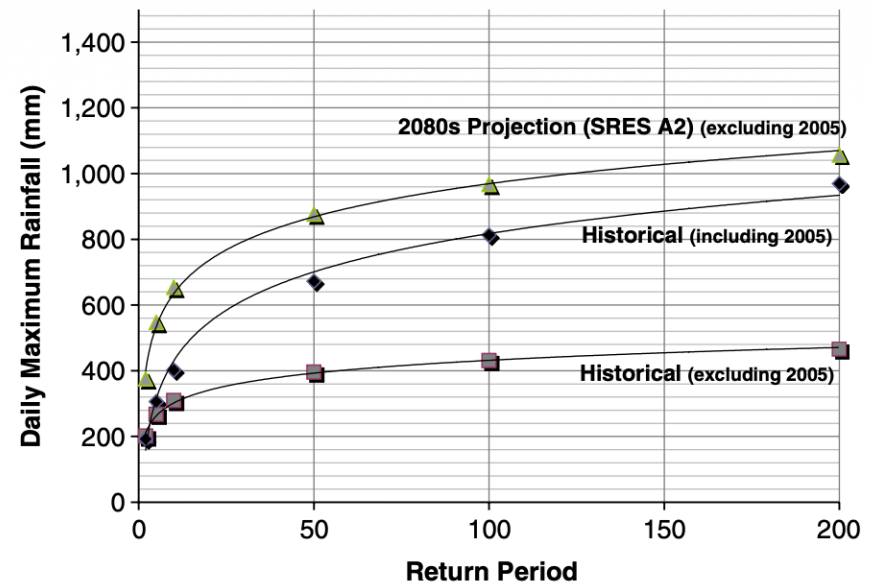
Reference: Ranger et al. (2011)



Mumbai: Flood Risk and Climate Change

By the 2080s, the upper-bound scenario suggested:

- more intense extreme rainfall
- more than double the likelihood of a 2005-like event
- sea-level rise reducing drainage effectiveness
- Reference: Ranger et al. (2011)



Return period: The estimated average number of years between rainfall events of the same or greater magnitude.

Mumbai: Adaptation Options

The study considered:

- restoring and widening rivers and channels
- improving the drainage system
- reducing the vulnerability of buildings
- strengthening insurance and recovery capacity

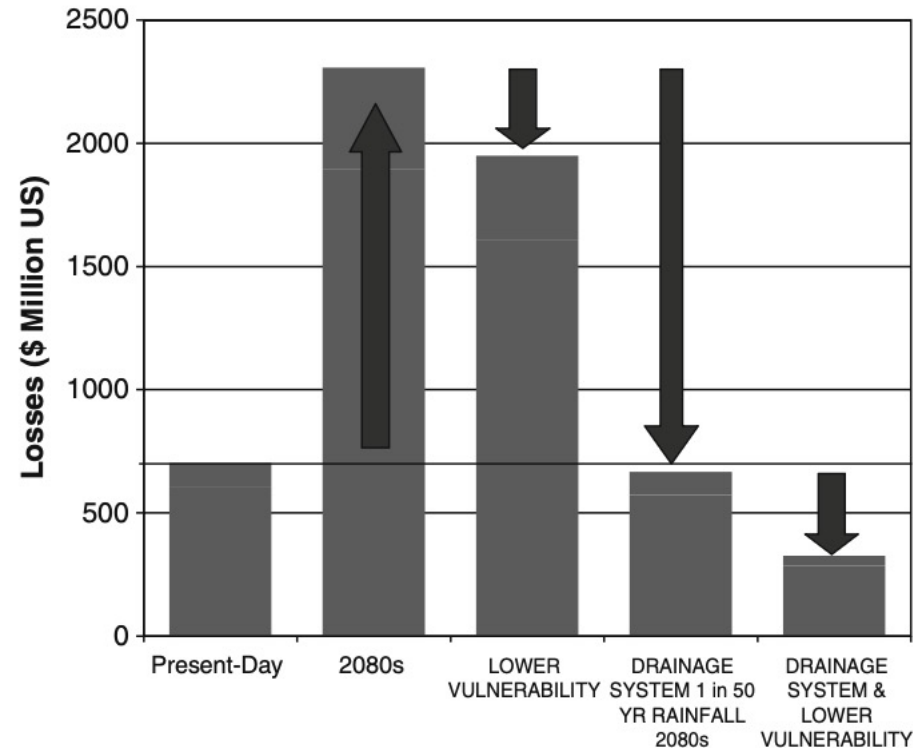
Different measures address different parts of risk.

Reference: Ranger et al. (2011)

Mumbai: Adaptation Reduces, but Does Not Remove Risk

Modelled drainage improvements could reduce losses from today's 1-in-100-year event by about 70%.

Reference: Ranger et al. (2011)



Estimated losses from a 1-in-100-year Mumbai flood under: (a) present-day conditions; (b) 2080s climate with no adaptation; (c) 2080s with building vulnerability reduced by 15%; (d) 2080s with drainage upgraded to manage a 1-in-50-year rainfall event; and (e) 2080s with both stronger buildings and improved drainage. Source: Ranger et al. (2011), Fig. 8.

Connecting the Two Cases to Climate Concepts

- Ahmedabad: extreme heat crosses health and housing thresholds.
- Mumbai: extreme rainfall exceeds drainage capacity.
- In both cities:
 1. Exposure and vulnerability shape the damage
 2. Adaptation changes the consequences, not the hazard
 3. Adaptation has costs, limits and residual risk
 4. Climate risk = hazard × exposure × vulnerability.